4.3 Practical Contributions

- **Open-source Qiskit implementation:** The proposed CS-HQR framework and all comparison methods are implemented using the Qiskit platform to ensure reproducibility and transparency of the experimental setup.
- **Simulation-based validation:** Due to the limited number of qubits and restricted access to current quantum hardware, execution on IBM Quantum devices is not feasible. Therefore, all experiments are conducted using quantum circuit simulators to analyze the behavior and resource requirements of the proposed method.
- **Medical imaging application:** The proposed framework is evaluated on Brain MRI images to demonstrate its potential applicability in medical image representation while maintaining perceptual image quality.

5 Problem Statement and Mathematical Formulation

5.1 Formal Problem Definition

Problem: Given a color image $\mathbf{I} \in \mathbb{R}^{N\times N\times 3}$ with RGB channels, design a quantum state preparation procedure $\mathcal{P}$ that encodes $\mathbf{I}$ into a quantum state $|\psi_I\rangle$ such that it supports efficient geometric operations like rotation and scaling while preserving color fidelity such that:

1. The quantum state $|\psi_I\rangle$ enables reconstruction $\hat{\mathbf{I}} = \mathcal{R}(|\psi_I\rangle)$ with $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$ where τ is a perceptual quality threshold
2. The preparation circuit $\mathcal{C}_\mathcal{P}$ minimizes the objective function:

$$\min_{\mathcal{P}} \left[\alpha \cdot Q(\mathcal{C}_\mathcal{P}) + \beta \cdot G(\mathcal{C}_\mathcal{P}) + \gamma \cdot D(\mathcal{C}_\mathcal{P})\right] \tag{22}$$

subject to $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$

where $Q(\cdot)$, $G(\cdot)$, $D(\cdot)$ represent the qubit count, total gate count, and circuit depth respectively, and $\alpha, \beta, \gamma \in [0, 1]$ are weighting coefficients satisfying $\alpha + \beta + \gamma = 1$.

5.2 Mathematical Formulation for Color Space Transformation

5.2.1 RGB to YCbCr Transformation

The RGB to YCbCr transformation translates an image from the RGB color space (red, green, blue channels) into YCbCr (luma + chroma channels). This splits brightness (Y) from color information (Cb, Cr), supporting more efficient compression and processing in video and image systems.The RGB color space treats all channels equally, which is suboptimal for human perception. We transform to YCbCr space where luminance (Y) and chrominance (Cb, Cr) are separated:

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \tag{23}$$

This transformation is derived from the ITU-R BT.601 standard and reflects the luminance sensitivity of the human visual system.

5.2.2 Inverse Transformation for Reconstruction

The inverse transformation recovers RGB values:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.402 \\ 1 & -0.344136 & -0.714136 \\ 1 & 1.772 & 0 \end{bmatrix} \begin{bmatrix} Y \\ C_b - 128 \\ C_r - 128 \end{bmatrix} \tag{24}$$

5.3 Chroma Subsampling Formulation

5.3.1 4:2:0 Subsampling Scheme

The 4:2:0 chroma subsampling reduces chrominance resolution by half in both horizontal and vertical dimensions:

$$C_b^{sub}(i,j) = \frac{1}{4}\sum_{m=0}^{1}\sum_{n=0}^{1} C_b(2i+m, 2j+n) \tag{25}$$

$$C_r^{sub}(i,j) = \frac{1}{4}\sum_{m=0}^{1}\sum_{n=0}^{1} C_r(2i+m, 2j+n) \tag{26}$$

For an $N \times N$ image:

- Luminance Y: $N \times N$ pixels (full resolution)
- Chrominance C_b^{sub}: $\frac{N}{2} \times \frac{N}{2}$ pixels
- Chrominance C_r^{sub}: $\frac{N}{2} \times \frac{N}{2}$ pixels

5.3.2 Data Reduction Analysis

For a 16×16 color image:

Table 3: 4:2:0 Chroma Subsampling for Reduced Data Representation

Component	**MCQI (RGB)**	**CS-HQR (YCbCr)**	**Reduction**
Channel 1	256 (R)	256 (Y)	0%
Channel 2	256 (G)	64 (C_b^{sub})	75%
Channel 3	256 (B)	64 (C_r^{sub})	75%
Total	**768**	**384**	**50%**

5.4 Quantum State Formulation

5.4.1 CS-HQR State Definition

The CS-HQR quantum state consists of two parallel components:

Circuit A - Luminance State:

$$|Y\rangle = \frac{1}{2^n}\sum_{i=0}^{2^{2n}-1} |Y_i^7 Y_i^6 \cdots Y_i^0\rangle \otimes |i\rangle \tag{27}$$

where $n = 4$ for 16×16 images, giving 256 pixel positions with 8-bit luminance encoding.

Circuit B - Chrominance State:

$$|CbCr\rangle = \frac{1}{2^{n-1}} \sum_{j=0}^{2^{2(n-1)}-1} \left(|C_j^7 \cdots C_j^0\rangle \otimes |j\rangle \otimes |ch\rangle\right) \tag{28}$$

where $|ch\rangle \in \{|0\rangle, |1\rangle\}$ selects between C_b and C_r channels, and j indexes the $\frac{N}{2} \times \frac{N}{2} = 64$ subsampled positions.

Combined CS-HQR State:

$$|I\rangle_{CS-HQR} = |Y\rangle \otimes |CbCr\rangle \tag{29}$$

Note: In practice, Circuits A and B execute in parallel on separate qubit registers or sequentially on the same register.This separation allows luminance and chrominance to be encoded with different spatial resolutions while preserving compatibility with NEQR-style deterministic retrieval

5.5 Complexity Analysis

5.5.1 Qubit Complexity

$$Q_{CS-HQR} = \max(Q_A, Q_B) = \max(2n + 8, 2(n-1) + 8 + 1) \tag{30}$$

For $n = 4$ (16×16 image):

$$Q_A = 8 + 8 = 16 \text{ qubits (position + color)} \tag{31}$$

$$Q_B = 6 + 8 + 1 = 15 \text{ qubits (position + color + channel)} \tag{32}$$

Therefore, $Q_{CS-HQR} = 16$ qubits (parallel execution).

5.5.2 Gate Complexity

For NEQR-style basis state encoding:

$$G_{NEQR}(N, q) = O(q \cdot N^2 \cdot H(I)) \tag{33}$$

where $H(I)$ is the average Hamming weight of pixel values.

For CS-HQR:

$$G_A = O(8 \cdot 256 \cdot H(Y)) \approx 2048 \cdot H(Y) \tag{34}$$

$$G_B = O(8 \cdot 128 \cdot H(C)) \approx 1024 \cdot H(C) \tag{35}$$

$$G_{CS-HQR} = G_A + G_B \approx 3072 \cdot \bar{H} \tag{36}$$

Compared to MCQI:

$$G_{MCQI} = O(24 \cdot 256 \cdot H(RGB)) \approx 6144 \cdot \bar{H} \tag{37}$$

Gate Reduction Ratio:

$$\eta_G = 1 - \frac{G_{CS-HQR}}{G_{MCQI}} = 1 - \frac{3072}{6144} = 50\% \tag{38}$$

5.5.3 Circuit Depth Analysis

Sequential encoding depth:

$$D_{CS-HQR} = D_A + D_B \approx \frac{G_A}{w_A} + \frac{G_B}{w_B} \tag{39}$$

where w_A, w_B are parallelization widths.

Parallel execution depth (with sufficient qubits):

$$D_{CS-HQR}^{parallel} = \max(D_A, D_B) \tag{40}$$

5.6 Perceptual Quality Guarantee

5.6.1 SSIM Analysis

Unlike PSNR or MSE, SSIM is aligned with human perceptual judgment, which directly motivates CS-HQR's design. Therefore, SSIM is not only the exact bit reconstruction but also the appropriate success criterion for NISQ-oriented quantum image representations.The Structural Similarity Index (SSIM) between original $\mathbf{I}$ and reconstructed $\hat{\mathbf{I}}$ is:

$$\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) = \frac{(2\mu_I\mu_{\hat{I}} + c_1)(2\sigma_{I\hat{I}} + c_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + c_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + c_2)} \tag{41}$$

For CS-HQR with bilinear interpolation of chrominance:

$$\text{SSIM}_{CS-HQR} \geq 0.98 \text{ (empirically validated)} \tag{42}$$

This threshold exceeds the just-noticeable-difference (JND) threshold for human perception, ensuring perceptually lossless quality.

6 Dataset Description

6.1 NIST-MNI Medical Imaging Dataset

Our experiments employ the NIST-MNI brain imaging dataset [41, 51], a standardized benchmark comprising 1,000+ T1-weighted MRI scans from healthy adult subjects. Curated by the Montreal Neurological Institute and calibrated by NIST for medical imaging benchmarks, this dataset supports rigorous assessment of quantum image representation fidelity under perceptual constraints.MRI scanners inherently produce single-channel intensity images.However, this characteristic is inherent to medical imaging modalities such as MRI, which are fundamentally intensity-based acquisition systems. In clinical imaging pipelines, grayscale MRI scans are frequently stored or visualized in RGB format for compatibility with standard imaging frameworks, resulting in nearly identical R, G, and B channels.Variance characteristics of luminance and chrominance channels in the MRI dataset. The chrominance channels exhibit very low variance due to the grayscale acquisition nature of MRI images. In future work, the proposed CS-HQR framework will be extended to support a wider range of high-definition color image datasets beyond the current experimental setup. Specifically, we plan to integrate multiple high-resolution color images and evaluate the scalability of the proposed quantum image representation technique for large-scale visual data. This includes adapting the encoding strategy to efficiently handle ultra-high-definition (UHD) images while maintaining reduced qubit utilization and optimized circuit depth. Additionally, further research will explore unified encoding schemes capable of merging diverse color image datasets within the same quantum representation framework, enabling broader applicability of the proposed technique in advanced quantum image processing and quantum machine learning tasks.

Table 4: Variance analysis of YCbCr channels for MRI dataset

Channel	**Variance**
Y	High
Cb	Very Low
Cr	Very Low

6.1.1 Dataset Characteristics

Table 5: NIST-MNI Dataset Specifications

Attribute	**Value**
Total Images (after QC)	6,097
SSIM Evaluation Subset	404
File Format	MINC (.mnc)
Original Resolution	Variable (typically 181×217×181)
Processed Resolution	16×16 pixels
Bit Depth	8-bit grayscale (converted to color)
Image Type	Brain MRI scans
Modality	T1-weighted structural MRI

6.1.2 Local Dataset Organization (`group4` Folder)

All experiments in this study use the locally organized dataset folder `group4/`, structured into 13 subject/group subfolders (`01`–`13`) containing MINC slice files under `2D/`. The raw `group4` directory contains 6,122 `.mnc` files; after preprocessing and quality-control filtering (removing blank or corrupted slices), 6,097 slices are retained for evaluation.

6.1.3 MINC File Format

The Medical Imaging NetCDF (MINC) format is a hierarchical data format designed for medical imaging applications. Key features include:

- Self-describing format with embedded metadata
- Support for multi-dimensional data (3D/4D volumes)
- Standardized coordinate systems for neuroimaging
- Lossless compression support

6.1.4 Preprocessing Pipeline

Each image undergoes the following preprocessing:

1. **Slice Extraction:** Extract 2D axial slices from 3D volumes

2. **Resizing:** Bilinear interpolation to 16×16 pixels
3. **Normalization:** Scale intensities to [0, 255] range
4. **Color Conversion:** Convert grayscale to RGB (for fair comparison with color methods)
5. **Quality Control:** Remove blank or corrupted slices
6. **Data Augmentation:** Apply simple augmentations such as small rotations (e.g., ±15°) and horizontal/vertical flipping where appropriate to increase sample diversity

6.2 Justification for Dataset Selection

Medical imaging provides an ideal test domain for CS-HQR because:

1. **Clinical Relevance:** Brain MRI is a high-value application where quantum speedups could provide tangible benefits
2. **Perceptual Requirements:** Diagnostic accuracy depends on preserving clinically relevant features, aligning with SSIM-based evaluation
3. **Standardization:** NIST-MNI provides reproducible benchmarking with established ground truth
4. **Image Characteristics:** Medical images exhibit spatial correlations that benefit from intelligent encoding strategies

6.2.1 Rationale for Grayscale-to-Color Conversion

A critical design decision in our experimental methodology is the conversion of originally grayscale medical images to RGB color space. This choice is motivated by **benchmarking fairness** rather than clinical necessity. Since CS-HQR is specifically designed for color image encoding and our primary comparison targets (MCQI, QRMW, EFRQI-color) are color-capable methods, evaluating all methods on identical color inputs ensures an equitable comparison. Simply using grayscale images would unfairly disadvantage color methods while artificially favoring grayscale-only encodings (NEQR, GQIR) that cannot scale to real-world color imaging scenarios. The grayscale-to-RGB conversion replicates intensity values across all three channels ($R = G = B = Y$), creating a controlled test case where the "ground truth" color information is perfectly correlated. This approach isolates the algorithmic efficiency of each encoding method from confounding factors such as natural color image statistics. Furthermore, modern medical imaging increasingly incorporates color information through pseudo-coloring, multi-modal fusion, and functional overlays, making color encoding capability clinically relevant for future applications.

7 Proposed Method: CS-HQR (Chroma Subsampling Hybrid Quantum Representation)

This section presents the complete CS-HQR (Chroma Subsampling Hybrid Quantum Representation) methodology, including the theoretical foundation, algorithm design, and circuit implementation.